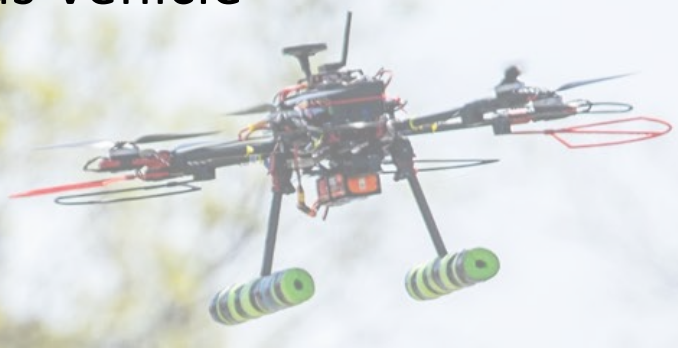
Raytheon
An RTX Business

‘2627 Raytheon Autonomous Vehicle Competition AVC Rules

Fall 2026 – Spring 2027



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1 INTRODUCTION

1.1 Mission & Vision

Welcome to the Annual Raytheon Autonomous Vehicle Competition!

Raytheon's mission is to partner with universities across 4 different regions to develop top talent, strengthen university relationships, and enhance the Raytheon mission.



The Raytheon Autonomous Vehicle Competition (AVC) is an event sponsored by Raytheon, an RTX business. The purpose of the AVC is to provide students an opportunity to solve existing real-world problems by using their creativity, innovation, and team-solving skills to design, develop, integrate, and test an Unmanned Vehicle system.

This team-oriented effort is an opportunity for university students to engage and put into practice their project-management and problem-solving skills on open-ended problems in a professional, industry work environment. For the corporate sponsor, it provides the opportunity to evaluate and assess top graduates and explore the latest technologies.





The Raytheon Autonomous Vehicle Competition will take place in the West, South, East, and Puerto Rico regions at an indoor venue over the course of 2 days. Specific universities will be revealed in a follow-on appendix.

Specific universities will be revealed in a follow-on Appendix to be distributed at a later date.

Note: If there is need for region specific guidance, rule clarification, challenge modifications, please refer to the Region-Specific Attachment accompanied by this Rules of Engagement to be released at a future date.

2 COMPETITION

2.1 Introduction

The Raytheon AVC challenges students with the design, implementation, integration, and test of complex system of systems. This requires developing skills in electromechanical design, embedded development, sensor systems, wireless communications, computer vision, cloud & edge processing, artificial intelligence, autonomous & avoidance algorithms, command and control systems, integration & test planning, and advanced manufacturing. Competition day is the team's opportunity to put all your skills and labor to the test.

The competition event is intended to be a fun, challenging, and high energy exercise. The goal of the event is to test your project and experience the satisfaction of having taken a system from a blank sheet of paper to a real-world implementation competing against your peers. The accomplishment of all the hard work throughout the year is an immense reward.

Note: Gameday decisions may be made to accommodate the competition's goals. Flexibility is key!

2.2 Theme: "Mission Bird Dog" - AI Integration, GPS Denied Navigation, Target Identification, Collaborative Autonomy

The AVC scenario for the '26-'27 season is dubbed "Mission Bird Dog"! Scenarios will include at least 1 vehicle provided by each team. The autonomous systems are not limited in quantity of vehicles or type. Each system can consist of multiple Unmanned Aerial Vehicles (UAV) and/ or multiple Unmanned Ground Vehicles (UGV).

The mission field will consist of several, different colored objects laid out in an indoor GPS denied environment. At the beginning of the mission, the team will be told which object the Autonomous Vehicle System needs to retrieve and bring back to the designated starting zone. All other objects on the mission field shall be avoided. **The team will communicate Leader's Intent with the Human-Machine Teamed Autonomous Vehicle System through natural language to retrieve the correct target object. The allowed form of communication is verbal or text (e.g. via microphone or chat).** The judges can change the target object for data retrieval during the exercise. These changes will be undisclosed prior to the competition. All processing is done at the edge.

The Autonomous Vehicle System is defined as one or more Autonomous Vehicles and the system that supports the autonomous vehicle(s).

Leader Intent Statement

Leader's Intent is a short, human and machine interpretable statement that explains the mission's purpose, what matters most, and what a successful outcome should look like. The intent gives the system enough information to act independently, adapt when conditions change and achieve the desired outcome without needing step-by-step instructions.

In this competition, Leader's Intent includes:

- **Purpose**: Which target object the system must retrieve and why that matters for the mission.
- **Priority Effects / Key Tasks**: Retrieve the correct target object, avoid all other objects, and navigate safely inside a GPS-denied indoor environment.
- **Constraints**: The system must stay within the allowed area, may only use edge processing, and must not touch or collect any ball other than the designated target.
- **Risk Tolerances**: The system should avoid unsafe paths, collisions, or actions that could damage itself, objects, or the environment.
- **Authorized Autonomy Envelope**: The level of freedom the system is allowed — it can choose its own route and movement strategy as long as it follows the intent and rules.
- **End State**: The correct target object is delivered back to the starting zone without any rule violations.
- **Handoff / Termination Criteria**: The mission ends successfully when the correct target object is returned or terminates early if a rule violating action occurs or the intent is superseded by a new one.

Teams provide Leader's Intent verbally or via text (e.g., microphone or chat). Judges may change the target object during the mission without prior notice, and the system must use the updated intent to immediately adjust its behavior. Leader's Intent allows the autonomous vehicle to remain goal aligned, safe, and effective even when unexpected changes occur.

Example Leaders Intent Statements (Note: box is used just as a placeholder for the target object type).

Below are similar statements phrased in different ways:

- "Retrieve the blue box and bring it back to the starting zone while avoiding all other objects."
- "Target has changed: return the yellow box to the start safely and ignore everything else."
- "Your goal is to collect the red box and navigate around any obstacles to deliver it to the starting zone."

- “Focus on getting the green box and avoiding contact with every other object on the field.”
- “New tasking: find and bring back the orange box without touching other objects or leaving the safe pathway.”

Other type of Purpose/Mission changes that could be requested by the Judges.

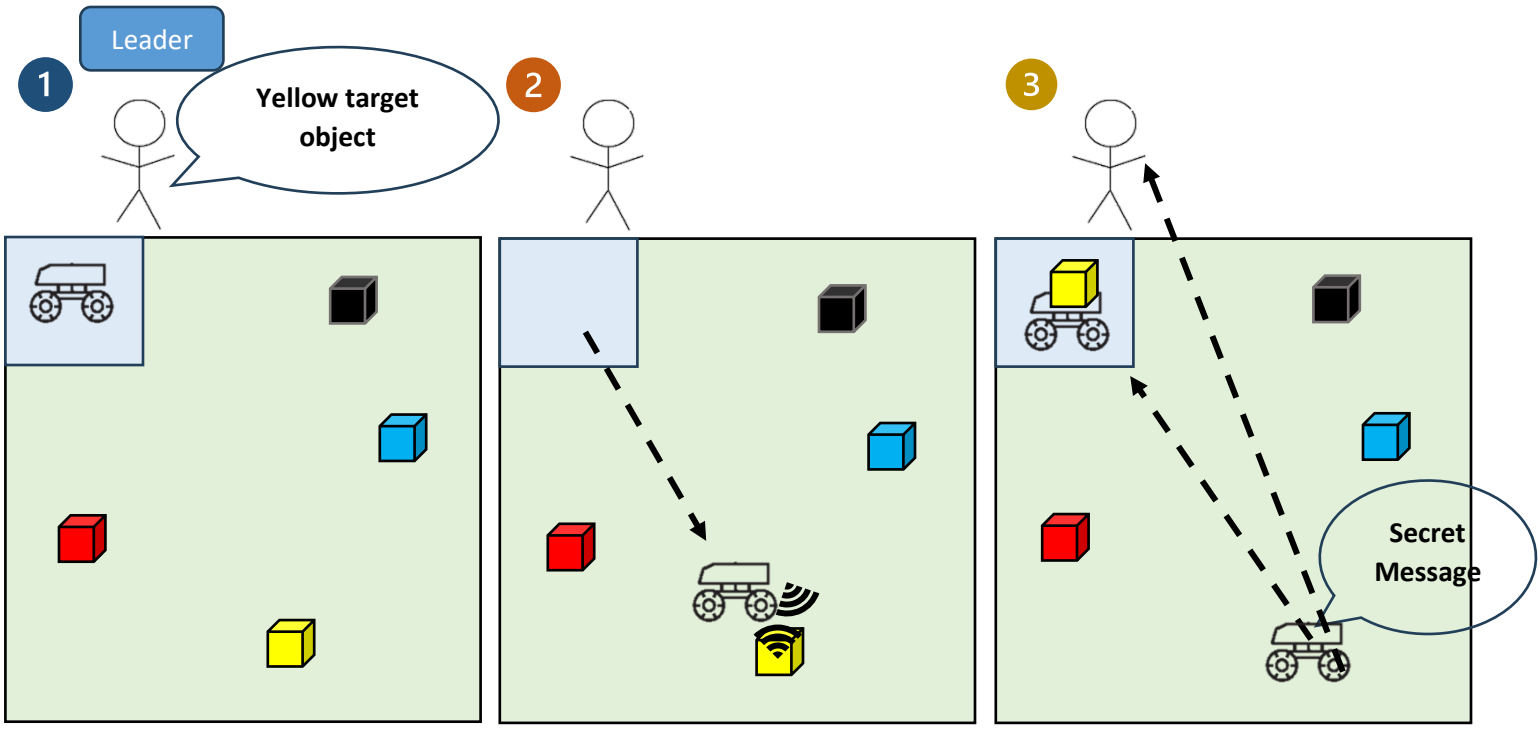
- “Abort the mission, remain undetected.”
- “Avoid the southwest quarter of the field.”
- “The blue object has been reported to have moved to the northeast quarter of the field.”
- “Stay to the far south of the field until you reach the eastern half and then obtain the blue object in the northeast corner.”

2.3 Challenge 1 – Base Bird Dog Challenge

The team will be told the target object to communicate Leader's Intent to the Autonomous Vehicle System. The AVS will navigate the mission field, avoiding non-target objects, to the target object. The AVS retrieves the Near Field Communication (NFC) message from the target object and sends it to leader. Bonus: the AVS also retrieves the object and returns it to the designated starting zone. Time limit: 7 minutes. Time stops once the leader receives the secret NFC message off the Target Object .

1. The AVS receives the Leader's Intent
2. The AVS navigates the mission field to the target object.
3. The AVS retrieves the secret NFC message from the target object and sends it to the leader. **Bonus**: the AVS also retrieves the object and returns it to the designated starting zone

The following pictures are an example of an AVS comprised of one 1 UGV searching for the target object and retrieving it back to the starting zone within 7 minutes. In this example the target object is a yellow box. NOTE: this is just an example and not to be interpreted as requiring a specific type of vehicle or the only acceptable form of verbal communication with the Vehicle System.

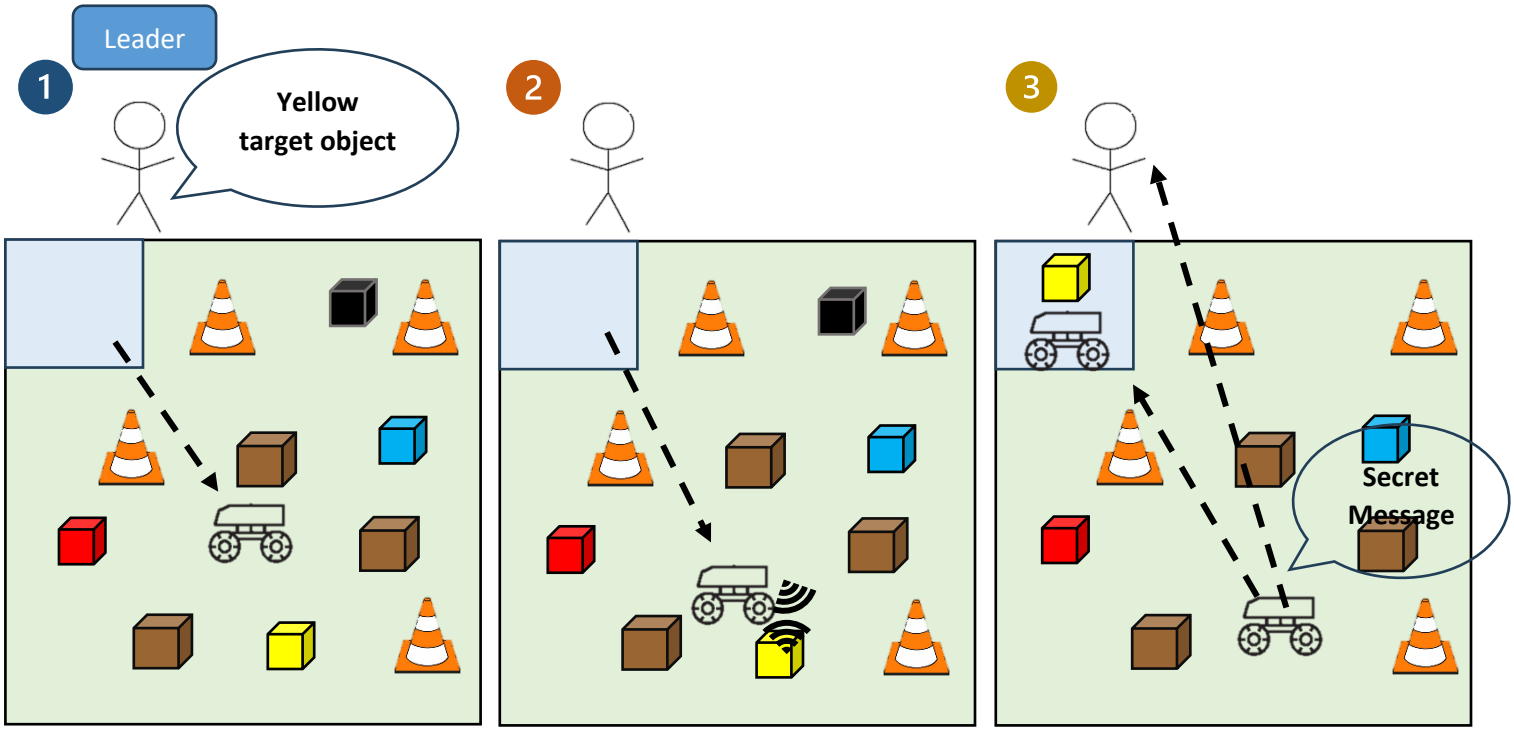


2.4 Challenge 2 – Congested Bird Dog Challenge

The team will be told the target object to communicate Leader's Intent to the Autonomous Vehicle System. The AVS will navigate the mission field, avoiding non-target objects, to the target object. The AVC retrieves the NFC message from the target object and sends it to leader. Bonus: the AVS also retrieves the object and returns it to the designated starting zone. Time limit: 7 minutes. Time stops once the leader receives the secret NFC message off the Target Object .

1. The AVS receives the Leader's Intent
2. The AVS navigates the mission with added objects field to the target object.
3. The AVS retrieves the secret NFC message from the target object and sends it to the leader. **Bonus:** the AVS also retrieves the object and returns it to the designated starting zone

The following pictures are an example of an AVS comprised of one 1 UGV searching for the target object and retrieving it back to the starting zone within 7 minutes. In this example the target object is a yellow box. NOTE: this is just an example and not to be interpreted as requiring a specific type of vehicle or the only acceptable form of verbal communication with the Vehicle System.

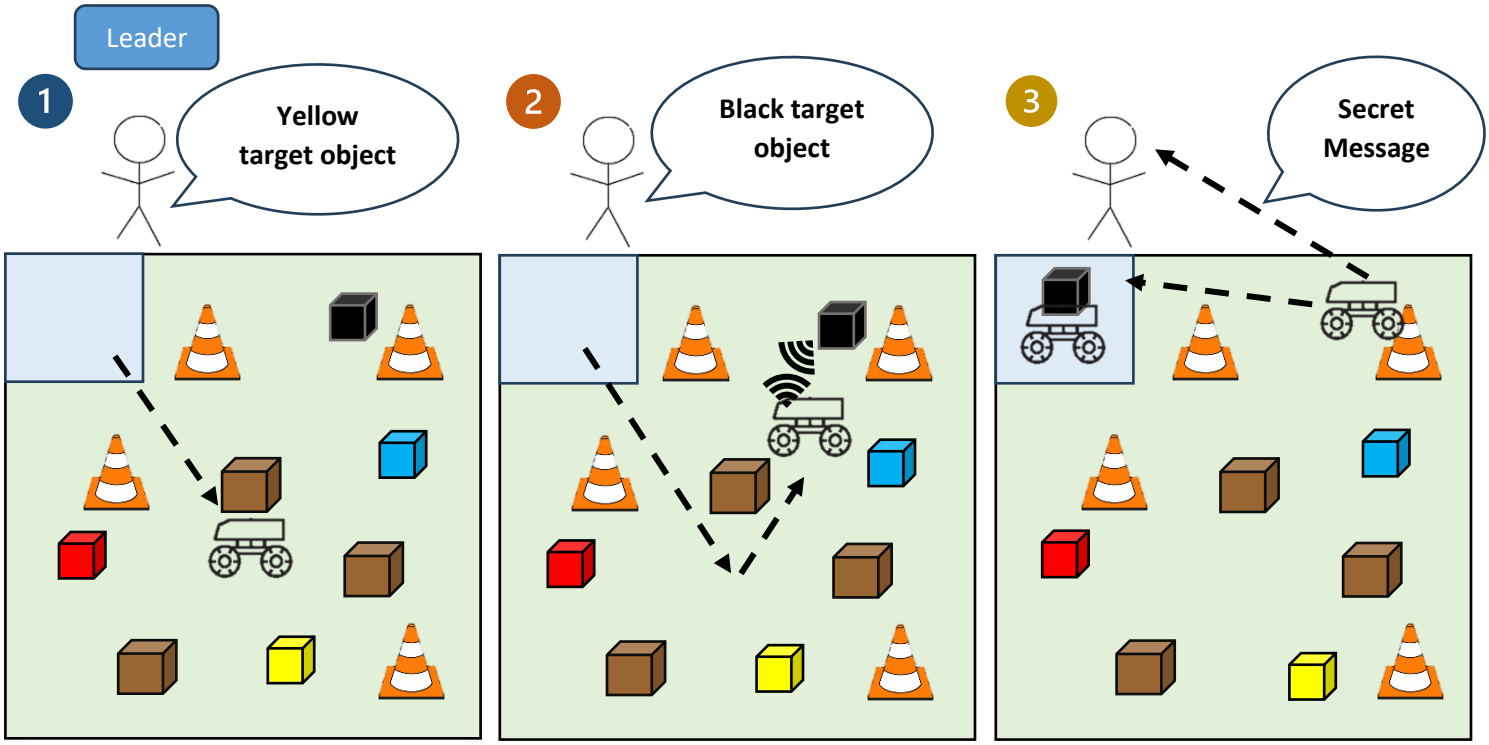


2.5 Challenge 3 – Replan Bird Dog

The team will be told the target object to communicate Leader's Intent to the Autonomous Vehicle System. The AVS will navigate the mission field, avoiding non-target objects, to the target object. At an undisclosed time, the target object will change. The AVS retrieves the NFC message from the target object and sends it to leader. Bonus: the AVS also retrieves the object and returns it to the designated starting zone. Time limit: 7 minutes. Time stops once the leader receives the secret NFC message off the Target Object.

1. The AVS receives the Leader's Intent and begins navigating the mission field.
2. The target object changes and the AVS adjusts navigation through the obstacles to the new target object
3. The AVS retrieves the secret NFC message from the new target object and returns it to the leader. **Bonus:** the AVC also retrieves the object and returns it to the designated starting zone.

The following pictures are an example of an AVS comprised of one 1 UGV searching for the target object and retrieving it back to the starting zone within 7 minutes. In this example the initial target object is a yellow box but changes to a black box. NOTE: this is just an example and not to be interpreted as requiring a specific type of vehicle or the only acceptable form of verbal communication with the Vehicle System.



2.6 Obstacles

Obstacles in Challenge 2 and 3 will be three dimmensional objects the UGV can not drive over. These will consist only of: small, medium or large sized cardboard boxes, traffic cones, and buckets. There will be an unknown amount of obstacles present but the minimum spacing between the objectacles will be 5 feet.

2.7 Target Objects

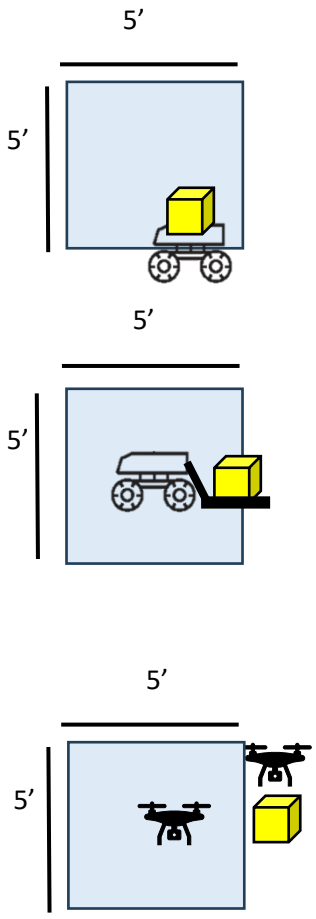
The Target Objects will consist of 6-inch foam dice from Amazon. There will be the following four colors: blue, red, yellow, black.

Link:
https://www.amazon.com/dp/B07C596DLZ?_encoding=UTF8&psc=1&ref_=cm_sw_r_cp_ud_dp_328GM8S5NS8BA6S3NP2J

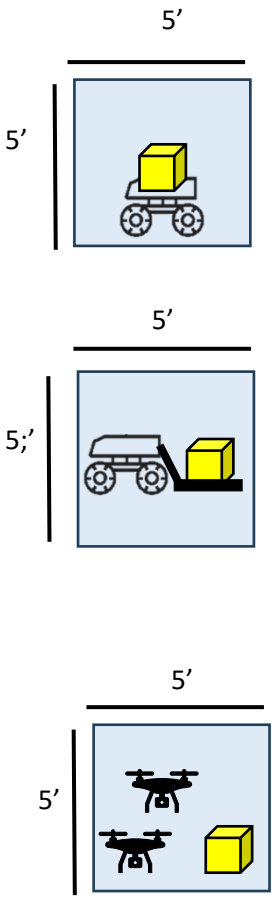
2.8 Starting Zone Requirements

The entire Vehicle System and the target object are required to return fully to the designated starting zone. The designated starting zone is defined as a 5-foot square located at the corner of the field the judges select. This will be taped off in blue painters' tape on competition day.

Does NOT count as returning to the starting zone



DOES count as arriving at the starting zone



2.9 Project Management

Students are expected to manage their project using industry's best practices. Students are responsible for project management training on their own time (e.g., Agile methodologies, DevSecOps, etc.). The university is expected to provide project management resources (e.g., MS Project, TEAMS, Jira, GitHub, MS Office, etc.)

Note: What has worked well in the past is for the teams to use a collaboration platform (e.g., TEAMS, Discord, Slack, etc.) and invite the Raytheon mentors, university faculty and SMEs, and any other indirect participants that would be able to help - as an example: any partner company technical POCs (as needed, i.e. BrainChip)" to that platform.

The Raytheon mentor will attend periodic team meetings as set by the team and mentor. Mentor meetings typically meet weekly, and last about 1 to 2 hours per week. Teams should meet as frequently as needed to manage the project with or without the Raytheon mentor.

3 RULES

3.1 General Rules

3.1.1 Budget

- Each university will have a baseline budget of $5,000 USD to use on this project. This budget can be used for hardware and software expenses to help the students to develop their prototype. **For 2027, there is no cap on the amount of money that can be used for the solution. A team is allowed to raise funds to support a more expensive solution.**
- Procurement of hardware and software will be done via the university procurement process.
    - Students will be asked to provide a budget breakdown prior to the Competition.
- The use of open-source hardware and software is recommended.
- **For teams desiring a standardized Unmanned Aerial Systema kit can be provided by SkyLabs Research and HardPoint Solutions for a TBD set cost.**
    - Kit requests can be made directly to SkyLabs Research or through the Raytheon Mentor.
- Schools can use resources from previous competition years to help prototype and test their solutions. Note: there is risk in using old platforms without proper background knowledge.

3.1.2 System Operation

- The system shall be capable of navigating around three-dimensional, real-life obstacles on the competition field and arrive at the prescribed destination
- **There is no limit to the number of autonomous vehicles to support a solution**

3.1.3 Field requirements

- 15 yards wide x 15 yard long (subject to change by venue)
- Consists of a flat, indoor surface such as concrete or hardwood. This is dependent on region.
- Indoors

3.2 BrainChip

BrainChip offers a Neuromorphic approach to AI compute, as their compute methods are inspired by the human brain (highly efficient compute) and utilize at memory-based compute. BrainChip's technology offers a paradigm shift by allowing AI compute typically implemented on server farms in the "Cloud", to now be implemented on a battery on the true edge.

Teams will be required to incorporate the provided BrainChips into their system

Resources found here:

(https://doc.brainchipinc.com/examples/general/plot_5_voc_yolo_detection.html#sphx-glr-examples-general-plot-5-voc-yolo-detection-py)Akida vision edge learning — Akida Examples documentation (https://doc.brainchipinc.com/examples/edge/plot_0_edge_learning_vision.html#sphx-glr-examples-edge-plot-0-edge-learning-vision-py) Here is the latest Rasp PI user guide: Akida-RaspberryPi_EvalKit_UG_111b (https://brainchip.com/wp-content/uploads/2022/06/Akida-RaspberryPi_EvalKit_UG_111b.pdf)

3.3 Unmanned Vehicle Requirements

Students can conduct a trade study on what hardware and software configuration are necessary to complete the assigned tasks and will have to assess with their mentors the best path forward. Items to consider are weight, speed, complexity, location of components, battery life, open-source vs custom design, etc. Note: This is not an exhaustive list of requirements. Test Documentation and Requirements Matrix should list other items inherently necessary to meet the challenge objectives.

3.3.1 UGV Requirements (only applicable if a UGV is employed)

- Shall employ a kill switch to terminate autonomous operation and halt motion in <3 seconds.
- Shall **not** employ GPS.

3.3.2 UAV Requirements (only applicable if UAV is employed)

- Shall have propeller guards installed.
- Shall implement a kill switch that overrides programmed or autonomous operation
- Shall **not** employ GPS
- Shall comply with FAA regulations if applicable

3.3.3 Autonomy

The Vehicle System shall have the ability to interpret natural language instructions integrating Large Language Models. There is no limit on the amount of human to system communication used. A few example implementations could be:

- A team could choose to only call out the target object and automating the remainder of the challenge
- A team could verbally navigate the system through the mission field calling out each task such as “drive forward, turn left, turn right, stop” etc.

Based on the Challenge descriptions above, the Autonomous Vehicles shall complete their tasks and be programmed with flexibility for items such as:

- Start and End location of challenge.
- Flight operational boundaries
- Change in the field dimensions
- Time Duration of challenge
- Target Object Changing

3.3.4 RF Spectrum Utilization

- Each UAV and UGV shall comply with FCC requirements for RF emissions.
- Each Team shall provide a preliminary spectrum management plan to their Raytheon mentor by January 31st, 2027, to include:
    - Frequency / channel
    - Bandwidth
    - Power
    - Contingency plan for cross-team interference

3.4 Scoring

Reference the Scoring Examples in Section 7 of this document for scoring specifics.

Note: Only after all challenge objectives have been met will time be used as a tie breaker. If the vehicle system and the correct target object does not fully return to the designated starting zone, if obstacles were hit, etc, the team will not receive points for fully meeting the objective and will not be eligible to beat teams that did meet the challenge objective.

3.5 Healthy Tips

Obtain training for:

- Project Management (PM)
- Agile (entire Team)
- SRR, SDR, CDR, TRR
- Machine Learning
- Image Recognition
- UAV Pilot

Order your Team T-Shirt Early!!!

4 SCHEDULE

The universities will compete in their respective region. The timeline for the competition will differ from Region to Region. Each schedule will be agreed upon per region.

Students will follow best practice development methodologies and meet with university and corporate mentors frequently (e.g., weekly) to ensure project progress.

At a high level, students will:

1. Conduct studies on existing state of the art solutions and work with stakeholders to gather all specifications and requirements for the project.

2. Generate project artifacts based on school requirements (e.g. proposals, design documentation, final presentations, etc.).

3. Design, Prototype, Implement, Test, and Demonstrate solution.

4. Compete in the 2026-2027 Regional Raytheon Autonomous Vehicle Competition

5. Prepare all classroom documentation requirements (i.e., poster, presentation, final report)

4.1 Milestones

4.1.1 Semester Based Milestones Template / Notional Schedule

Note: this is a template, and not *required* for the project. It is meant to be a reference for building a schedule.

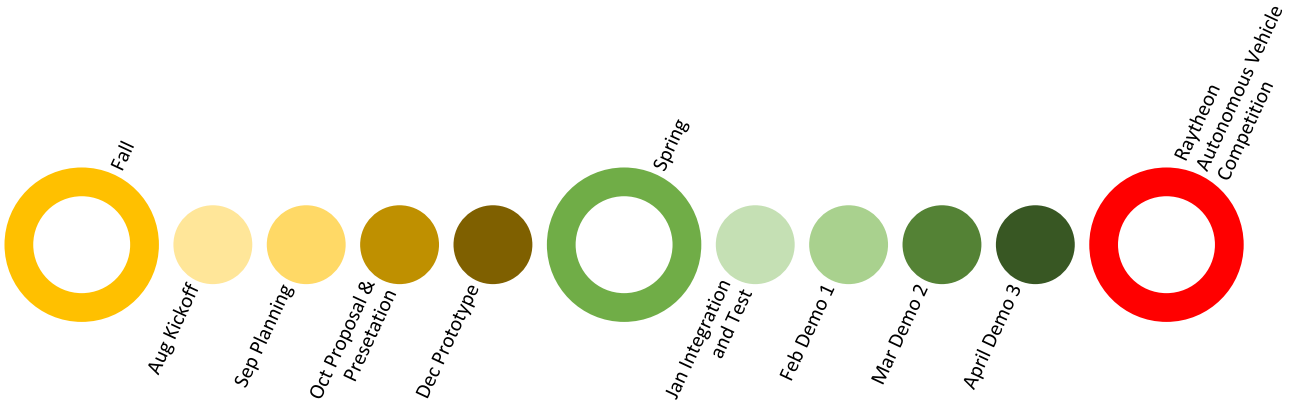


Figure 1 Semester Based Notional Milestone Template Set

Date	Milestones
August - Project Kickoff (August)	Requirements Review & Clarification / Q&A by students over the Rules of Engagement.
September - Project Planning (e.g., System Requirements Review)	Configuration Management Project Logistics
October	Project Proposal & Presentation (e.g., System Design Review) Order Parts
December	Prototype creation a. Mid-Year report is due (e.g., Critical Design Review) i. template can be provided if needed. b. Prototype Demonstration(s)
Spring Demos (e.g., Test Readiness Reviews)	a. Integration and Test b. AVC Demo 1 c. AVC Demo 2 d. AVC Demo 3
April	Raytheon Autonomous Vehicle Competition!

4.1.2 Quarter Base Milestone Template / Notional Schedule

Note: this is a template, and not *required* for the project. It is meant to be a reference for building a schedule.

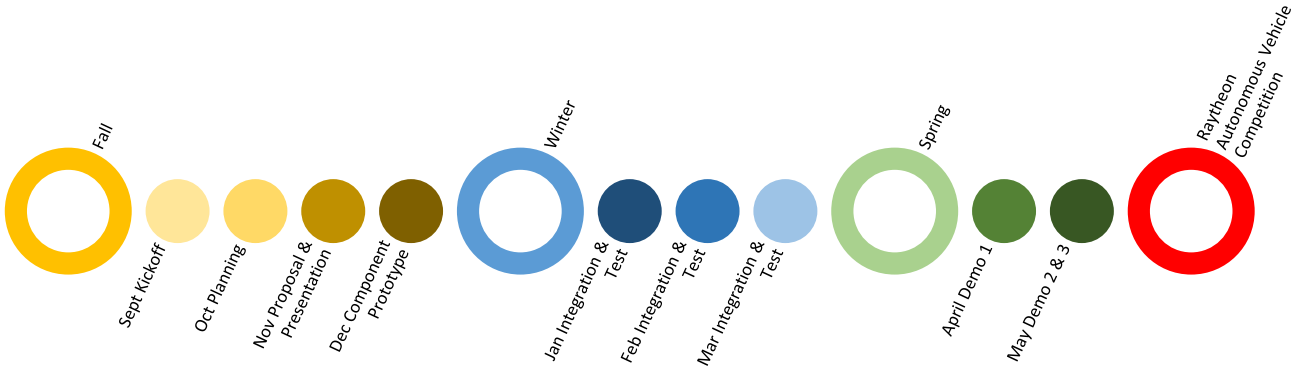


Figure 2 Semester Based Notional Milestone Template Set

Date	Milestones
September - Project Kickoff (August)	Requirements Review & Clarification / Q&A by students over the Rules of Engagement.
October - Project Planning (e.g., System Requirements Review)	Configuration Management Project Logistics
November	Project Proposal & Presentation (e.g., System Design Review) Order Parts
December - Component Prototype creation	Critical Design Review Component Prototype Demonstration(s)
January (e.g., Test Readiness Reviews)	Integration & Test
February	Integration & Test
March	Integration & Test
April	Demo 1
May	Demo 2 & 3
June	Raytheon Autonomous Vehicle Competition!

4.1.3 Competition Event Schedules by Region

AVC Regions	AVC Dates
Southern	
East Coast	
Puerto Rico	
West Coast	

5 DESIGN AND TEST GUIDENCE

As students design their system, they should follow standard engineering processes. University classes should have assignments guiding students through this process. This section will outline some important steps and pose questions to assist students in making design choices and testing their system.

Understand the Problem

Read the Rules Document thoroughly

Once provided, read the Qual Test thoroughly

Create a Requirements Matrix

Make a Plan and Assign Team Roles

Begin to ask questions and make high level design choices

- What subsystems can the overall deliverable be broken into?
- What research is required for each subsystem?
- What are the single point failure elements of the system?
- What software languages will be used?

Who will be doing what work?

Make a timeline with specific dates

Begin Designing in Detail and Order Parts

Ask more specific questions that will drive part selection

- If a UGV, how many and what type of wheels/tracks will be used? Can it drive several surfaces?
- What material can we use from previous years' designs?
- What parts will be needed to identify the Destination?
- How will the drones know their own location and height?
- How will all the hardware be mounted together?
- How will the hardware manage power?
- Can we build custom parts to make our lives easier and design cheaper? 3D Printing etc.
- Are the parts I want from approved vendors?

Begin making models, drawings and electrical interconnects for the selected parts

Prototyping and Testing

As designs become prototypes, begin testing your design while there is plenty of time to make changes. Review the Rules Document and ensure the design meets all the requirements. Continue asking questions to look for fatal design flaws. Your goal should be to find ways your system could fail and then go fix them!

- Will I have all my parts in time?
- Do I have a clear path forward to integrate all system elements.
- Is my design flexible?
- Do I have a way to power on everything at the same time?
- Is there anything that I'm over engineering?
- Did I code in a way that can be changed quickly on competition day? Think field size, speed, etc.
- How can I test and prove my system meets the requirements?
- How can I test that my system is reliable in different conditions?
    - How flexible are my cameras if applicable? Are they reliable at non-ideal angles? Different lighting?
- Document testing and cross reference with your Requirements Matrix!

Competition Preparation

Get excited for the big day as you enter the last phase of integration.

- **COMPLETE THE QUAL TEST AS EARLY AS POSSIBLE**
- Do I have tools and spare parts to perform drone surgery at the competition?
- Do I have a monitor, mouse and keyboard to easily debug coding errors that may appear at the competition?
- Read through the Rules again and make sure your system meets all of them. Requirements Matrix and Testing Documentation?
- Think of every possible way something could go wrong and then make a plan for how your system will overcome that!

5.1 Requirements Matrix

It is required to build a Requirements Matrix to help keep students organized throughout the duration of the project. It is a tool that ensures students do not forget requirements they have been given as well as preventing unnecessary scope creep. A Requirements Matrix is common practice for industry Factory Acceptance Tests. A sample of a partial Requirements Matrix can be seen below:

Requirement ID	Description
1	UAV shall have propeller guards.
2	UAV shall have a kill switch
3	UGV shall have a kill switch
4	System can display UAV velocity

5.2 Test Documentation

It is required for teams to produce Test Documentation. Testing Documentation ensures that every requirement has been met. It can also prevent unnecessary testing when a requirement has already been proved in a previous test. Writing and executing good tests will aid in overall organization and efficiency towards the end of the project. Thorough testing documentation is common practice for industry Factory Acceptance Tests. A sample of a partial Test Document can be seen below:

Test ID	Test Description	Requirements Tested
1	Verify the UAV has propeller guards installed	1
2	Autonomously launch the UAV and execute a straight flight path. Verify the system can display the UAV's velocity. In the middle of the flight, flip the kill switch. Verify the UAV safely stops flight and lands within 10 seconds of the kill switch being flipped.	2, 4
3	Autonomously start the UGV and execute a straight navigation path. In the middle of the path, flip the kill switch. Verify the UGV safely stops within 10 seconds of the kill switch being flipped.	3

6 ROLES AND RESPONSIBILITIES

6.1 Project Mentor

Each university will be given a Raytheon representative as a project mentor to help guide the project.

6.2 Judges

Judges will observe the final competition and grade based on performance and requirements met. Judges will not be project mentors.

6.3 Faculty Advisor

The Faculty Advisor(s) are the professors of each respective engineering domain. Faculty advisors are typically responsible for team creation, grades, monetary logistics, and course work.

6.4 Project Manager

Students can choose a PM that will lead all the teams and be responsible for team communication, logistics, requirements verification and validation, and overall success of the project.

6.5 Student Teams

Teams will consist of undergraduate students in the field of computer science and engineering disciplines.

7 AWARDS AND EVALUATION

Raytheon's desire is to encourage students to embrace innovation and real-world experience to further both the school and the individual's experience. Independent Raytheon leadership will evaluate each team's project based on the following categories:

Core Scored Elements:

Description	Weights
Requirements Matrix	5% Points for turning in a Requirements Matrix.
Test Documentation	5% Points for turning in Testing Documentation.
Qualifier Test	15 % Points for completing the qualifier test as pass/fail activity submitted via YouTube
Report / Video	10% Points based on the RTX evaluation presentation artifacts: Report / Video
Challenge 1	10% of execution of Challenge 1.
Challenge 2	20% of execution of Challenge 2.
Challenge 3	35% of execution of Challenge 3

Note: Only after all challenge objectives have been met will time be used as a tie breaker.

Bonus Opportunities:

Description	Weights
Early Bird Incentive Bonus	Tiered bonus for completing qualifier test early. Assume N is the number of teams on region, the first team to complete all the qual test receives (N-1) % bonus, second team (N-2)% and continue until the last team gets nothing. **Only Qual Test received a week before the Competition Date are eligible for these bonus points**. Idea is to incentivize experimentation and test earlier
STEM Outreach Bonus	**5**% bonus to be awarded to each team that completes a STEM outreach activity to school (any level). Example: Teams present their project to a local high school club.
High School Shadow	**5%** bonus to teams who mentor a high school student through a shadowing experience. The high school student should not take ownership of any project tasks but would gain experience through periodically shadowing the team throughout the project. Note, additional training may be required. Must begin shadowing by January of 2027 to receive full credit. Partial may be offered if started later.

Team Shirts	**1%** bonus to be awarded to schools who design and wear their own unique T-Shirts on Competition Day

Penalties:

Description	Weights
Calibration Points	5 Points will be deducted from the final score for each Calibration Point used on the field. Calibration Point is defined as a physical object added to the field for visual purposes to assist the system in special awareness of the field. Note this does not apply to a visual reference used on the UGV. Calibration points do not send off any kind of signal. Example: Red Dot on a corner of the field.
Beacon	10 Points will be deducted from the final score for each beacon device used. This includes dropping a Beacon on the destination or using a Beacons to map the field. Beacon is defined as any device outputting a signal to call out a location. Examples: Bluetooth, wifi, sound, radio, etc.

Scoring Examples:

If a Team in a region of 10 schools receives full credit for all the Core Scoring Elements and receives the max number of Bonus Points with no penalties, they would be scored as shown below with the percentage breakdown in red and the points scored in blue:

Core Scored Elements Score:

Qual Test Req Matrix Test Docs Report/Video C1 C2 C3

.15(100) + .05(100) + .05(100) + .1(100) + .1(100) + .2(100) + .35(100) = 100 Points

Bonus Points Opportunities:

Early Bird STEM Outreach Shadow T-Shirts

(10-1) + .05(100) + .05(100) + .01(100) = 20 Bonus

Total

Core Scored Elements Points + Bonus Points

100 + 20 = 120 Total Points

Scenario A: Team A has received all the available points on their Qual Test, Requirements Matrix and Test Documentation. However, they were not eligible for the Early Bird Bonus. They received full credit for their poster and video. On competition day they successfully completed both Challenge 1, Challenge 2, and Challenge 3 with no problems and received full credit. They did NOT complete a STEM outreach or have a shadow student, but they did make some super cool custom T-Shirts! They received a penalty for using one Beacon. They would be scored as follows with the percentage breakdown in red and the points scored in blue:

Core Scored Elements Score:

Qual Test Req Matrix Test Docs Report/Video C1 C2 C3

.15(100) + .05(100) + .05(100) + .1(100) + .1(100) + .2(100) + .35(100) = 100 Points

Bonus Points Opportunities:

Early Bird STEM Outreach Shadow T-Shirts

(0) + .05(0) + .05(0) + .01(100) = 1 Bonus

Total

Core Scored Elements Points + Bonus Points - Penalties

100 + 1 - 10 = 91 Total Points

Scenario B: Team B was the third school in a region of 10 to turn in their qual test early and received 132 points on their Qual Test. They did not complete a Requirements Matrix or Testing Documentation. They received full credit for their poster and video. On competition day they successfully completed Challenge 1 and Challenge 2 but only received 80/100 points for Challenge 3. They completed a STEM Outreach, had a shadow student and make T-Shirts. They received no penalties. They would be scored as follows with the percentage breakdown in red and the points scored in blue:

Core Scored Elements Score:

Qual Test Req Matrix Test Docs Report/Video C1 C2 C3

.15(100) + .05(0) + .05(0) + .1(100) + .1(100) + .2(100) + .35(80) = 73 Points

Bonus Points Opportunities:

Early Bird STEM Outreach Shadow T-Shirts

(10-3) + .05(100) + .05(100) + .01(100) = 18 Bonus

Total

Core Scored Elements Points + Bonus Points

73 + 18 = 91 Total Points

8 Appendix A – Internet Access (optional)

Internet access to the aerial and ground vehicles is optional; however, it may help facilitate flight and target identification options in appendices B and C.

8.1 Option 1: Local “Guest” network to "base station area"

If the supporting venue provides guest Wi-Fi to the staging area(s), any network usage passes through each team's base station to their vehicle(s) through telemetry radios. Most base station software (Mission Planner, QGC, etc.) support this functionality. Please note that host site’s networks may be unreliable both in connectivity and speed.

8.2 Option 2: "You're on your own"

In practice, this means bring your own cellular model and/or hotspot and infrastructure to ensure each component has the necessary Internet connectivity. It is recommended you test your AVS with your own internet at maximum load (e.g. with all end points) ahead of time at test events PRIOR to competition day, to ensure it has the speed, bandwidth, and reliability you will need on competition day."

9 Appendix B – Logging

Logging is required for all Autonomous Vehicles to help determine intended actions.

The following actions must be logged:

- Vehicle ID
- Vehicle System Start Time
- Destination discovery
- Communication between Leader and Vehicles
- Vehicle System End time

10 Appendix D Sky Labs Research

Sky Labs Research is a company created by students who have competed in the Raytheon Autonomous Vehicle Competition previously. They are offering consultation services to provide recommendations on drone parts and systems integration. The company / service is not affiliated with Raytheon in any way but is provided here as an example for teams who would like to try and accelerate their research and prototyping.

- Consultation services: https://skylabsresearch.com/products/consultations
- On the site, Sky Labs will be offering consultation meeting scheduling for students to meet and discuss how they can best be supported by Sky Labs. After the first meeting the service will transition to a free email correspondence, or you may schedule more meetings as needed.

Sky Labs will not provide design decisions/ideas but will help try and guide them to reach their own design choices.

- Free Nugget: eCalc - https://www.ecalc.ch/ is used to calculate drone power trains. It should be used by students during the drones' design phase.

11 Appendix E Discord

Please ask your mentor about joining the Discord with BrainChip for assistance using their products.